ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P2H Worked Solutions

Jan 2022 | P2H

33 questions ▪ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

TOPIC: **Expanding Brackets**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 1 Marks: 4

1 (a) Expand and simplify $(y + 4)(2 - y)$

....

(2)

(b) Factorise fully $15b^5c - 35b^3c^9$

....

(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Expanding Brackets**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

Jan 2022 P2H - Jan 2022 | P2H

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jan 2022 P2H - Jan 2022 | P2H

3

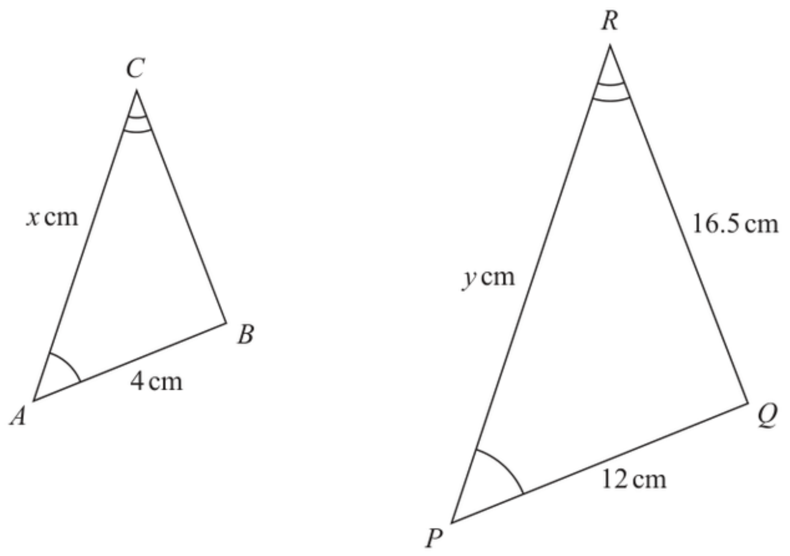

Diagram **NOT** accurately drawn

Triangle ABC is similar to triangle PQR

$AB = 4$ cm $PQ = 12$ cm $RQ = 16.5$ cm $AC = x$ cm $PR = y$ cm

(a) Calculate the length of BC

.... cm

(2)

(b) Write down an expression for y in terms of x

$y =$ ....

(1)

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 4 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P2H - Jan 2022 | P2H

4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

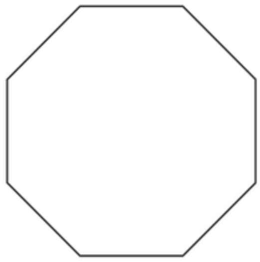

Diagram **NOT** accurately drawn

(a) Write down the lower bound of the length of each side of the octagon.

.... mm

(1)

(b) Write down the upper bound of the length of each side of the octagon.

.... mm

(1)

(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P2H - Jan 2022 | P2H

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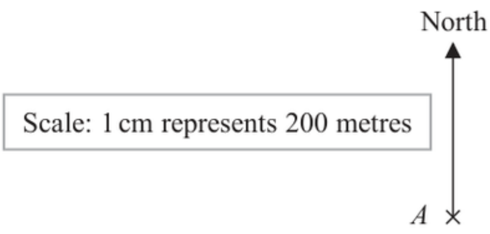
No worked solution is saved yet.

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2022 P2H - Jan 2022 | P2H

5 The scale diagram shows the position on a map of a house, A



House C is on a bearing of 110° from A
The distance from A to C is 700 m

(a) Mark the position of C on the diagram with a cross (×)
Label your cross C

(3)

(b) Write the scale of the map in the form $1:n$

1 : ....

(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 6 Marks: 5

TOPIC: **Probability Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

6 A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

Sweet	pink	white	green	red
Probability			0.2	0.35

The ratio
number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

(Total for Question 6 is 5 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 5

TOPIC: **Probability Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

TOPIC: **Prime Factors, HCF & LCM**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 7 Marks: 3

7 Find the lowest common multiple (LCM) of 28, 42 and 63
Show your working clearly.

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 8 Marks: 5

TOPIC: **Percentages**

Jan 2022 P2H - Jan 2022 | P2H

8 The table gives information about the average house price in England in 2018 and in 2019

Year	**2017**	**2018**	**2019**
Average house price (£)		228 314	231 776

(a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

.... %
(2)

The average house price in 2019 was 7.7% greater than the average house price in 2017

(b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£ ....
(3)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Percentages**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 9 Marks: 4

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

9 The frequency table gives information about the number of points scored by a player.

Number of points	**Frequency**
0	13
1	17
2	8
3	x
4	11

The mean number of points scored is 2

Work out the value of x

$x =$ ....

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

TOPIC: **Simultaneous Equations**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 10 Marks: 3

10 Solve the simultaneous equations

$$3x + 5y = 3.1$$
$$6x + 3y = 3.75$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 11 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2022 P2H - Jan 2022 | P2H

11 The diagram shows a regular 10-sided polygon, *ABCDEFGHIJ*

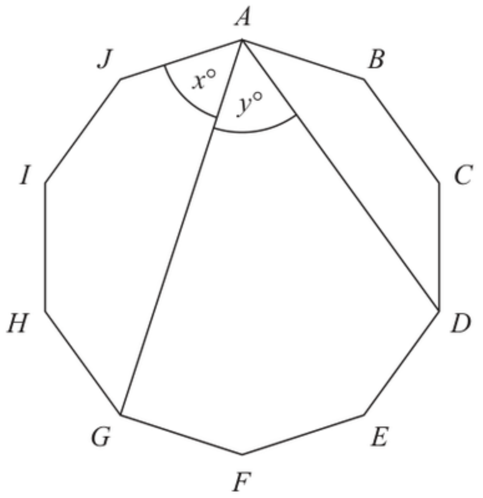

Diagram **NOT** accurately drawn

Show that $x = y$

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P2H - Jan 2022 | P2H

12 $a = 6 \times 10^{40}$

Work out the value of a^3
Give your answer in standard form.

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P2H - Jan 2022 | P2H

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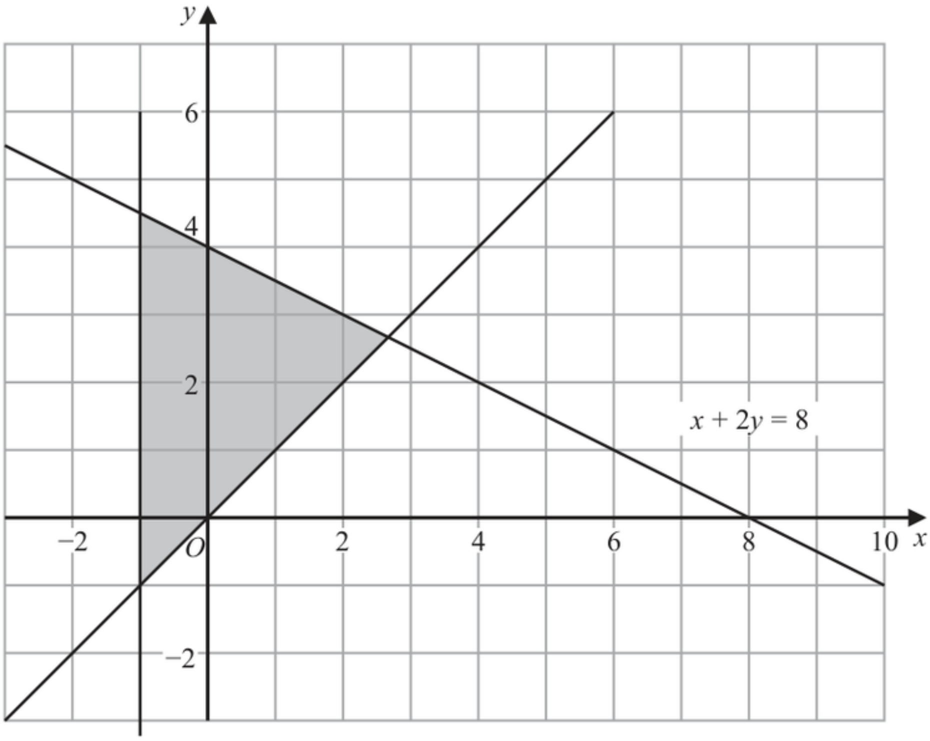
No worked solution is saved yet.

PAST PAPER QUESTION Q#: 13 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

13 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

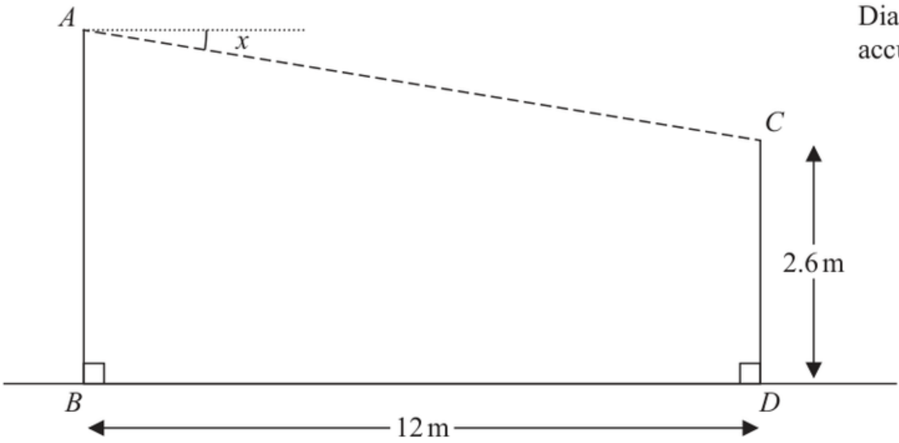
Q#: 14 **Marks: 3**

Topic: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2H - Jan 2022 | P2H

14 A zip wire is shown as the dashed line AC in the diagram.

Diagram **NOT** accurately drawn



The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$$CD = 2.6\,\text{m} \qquad BD = 12\,\text{m}$$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

.... m

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2H - Jan 2022 | P2H

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PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

15 Diyar recorded the distance, in kilometres, that he cycled each day for 11 days.
Here are his results.

8 10 12 13 5 23 21 7 5 16 14

Find the interquartile range of his results.

.... km

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

16 *D*, *E*, *F* and *G* are points on a circle, centre *O*

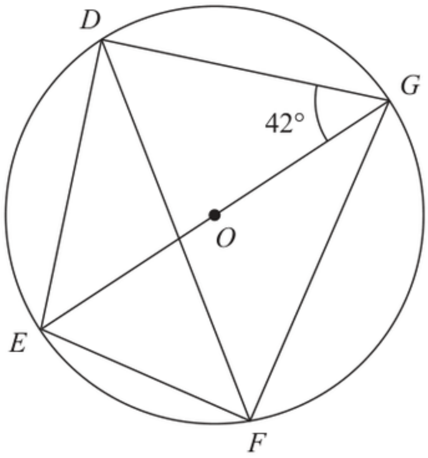

Diagram **NOT** accurately drawn

EOG is a diameter of the circle.

Angle *EGD* = 42°

Calculate the size of angle *DFG*
Give a reason for each stage of your working.

Angle *DFG* = ... °

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

17 Show that $\dfrac{\sqrt{12}}{\sqrt{3}+2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

Q#: 18 Marks: 3

18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2H - Jan 2022 | P2H

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PAST PAPER QUESTION

Q#: 19 Marks: 3

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

19

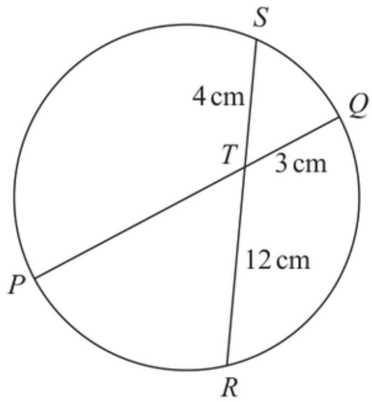

Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
RTS is a chord of the circle.

$TQ = 3$ cm $\qquad ST = 4$ cm $\qquad TR = 12$ cm

Calculate the radius of the circle.

.... cm

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Circle Theorems**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

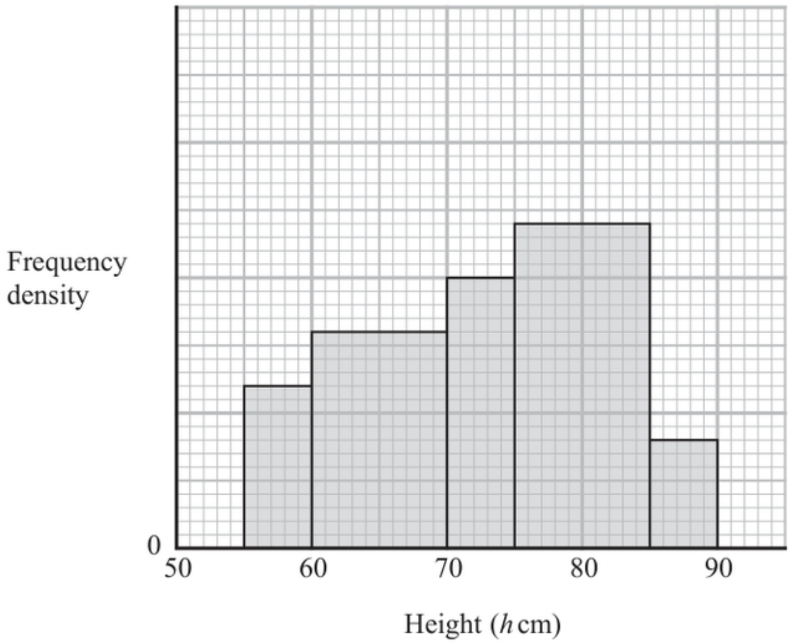
PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

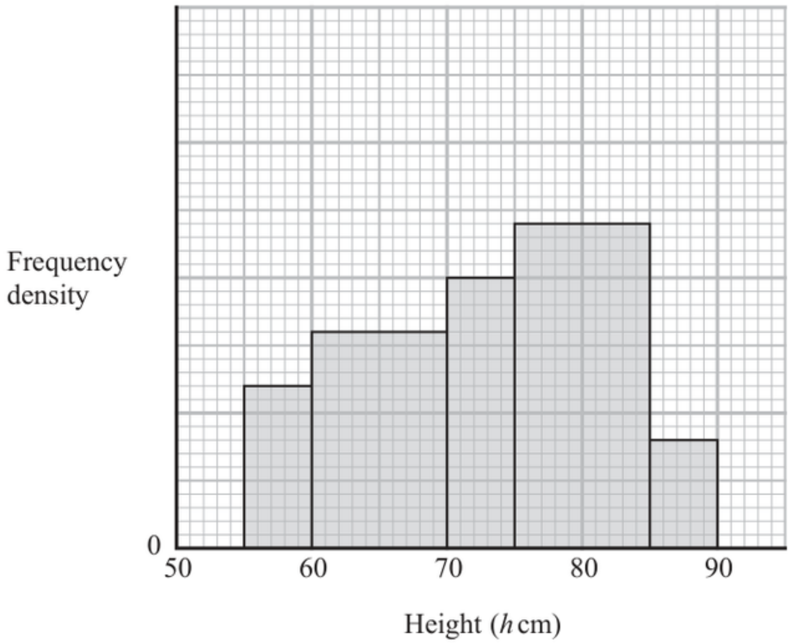
PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



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One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Histograms**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

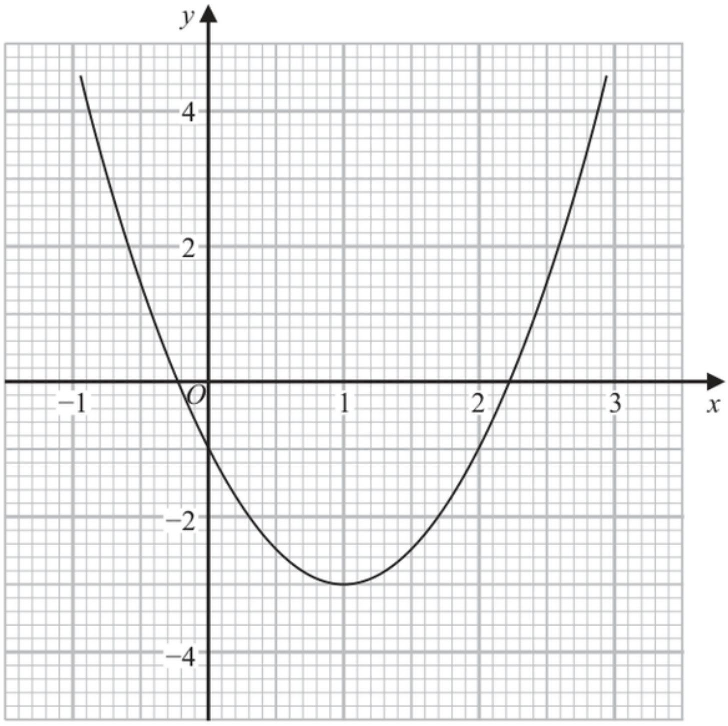
PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

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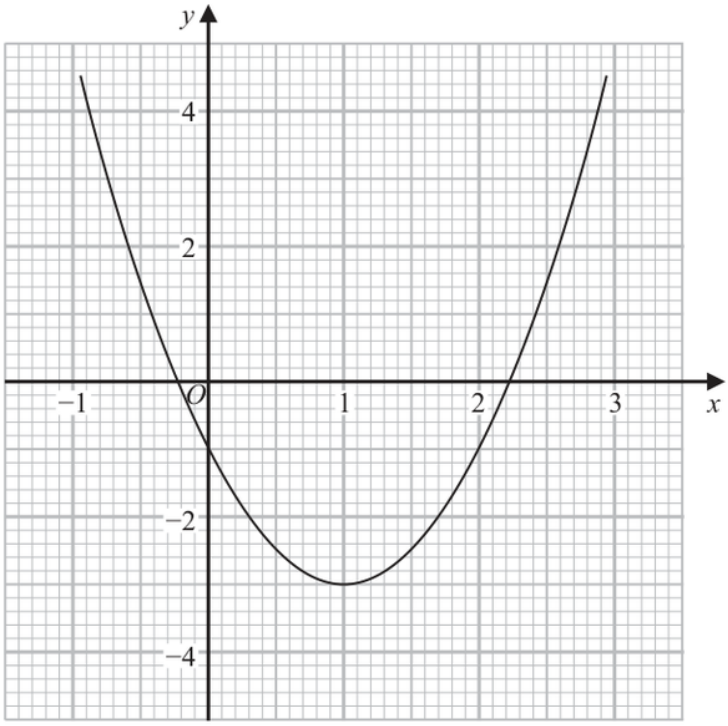

(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Graphs of Functions**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

22 Here is a rectangle.

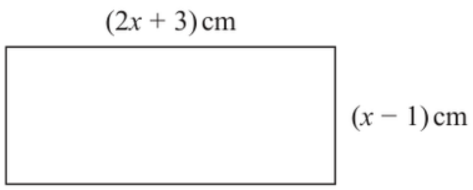

Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

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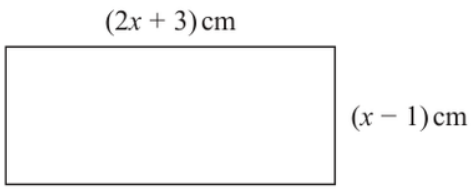

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(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Solving Inequalities**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

23 The diagram shows triangle PQR

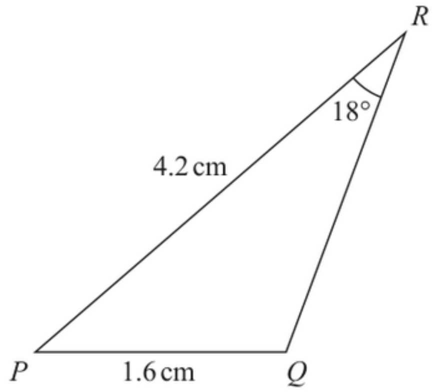

Diagram **NOT** accurately drawn

$PQ = 1.6$ cm $\quad PR = 4.2$ cm $\quad$ Angle $PRQ = 18°$

Given that angle PQR is obtuse,

work out the area of triangle PQR
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

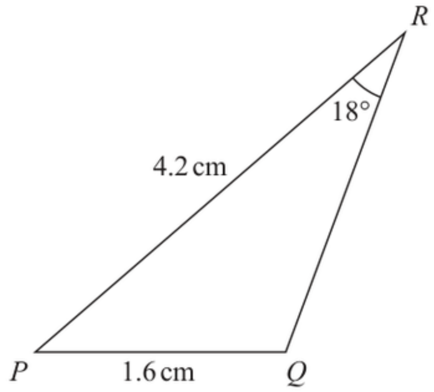
PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

23 The diagram shows triangle PQR



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Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geqslant 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

$a =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

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The acceleration of P at the instant when P is at A is $a\ \text{m/s}^2$

Find the value of a

$a =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Differentiation**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

25 The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \ldots$

$g^{-1} : x \mapsto$ ....

(4)

(b) State the domain of g^{-1}

....

(1)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

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$g^{-1} : x \mapsto$ ....

(4)

(b) State the domain of g^{-1}

....

(1)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Functions**

Jan 2022 P2H - Jan 2022 | P2H

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

26 An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$
$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$d =$ ....

$m =$ ....

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Jan 2022 P2H - Jan 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

PAST PAPER QUESTION

TOPIC: **Sequences**

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